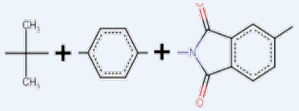
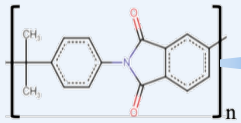
# Pre-training of Poly4mer:

## a. Physics in the data

1. **Sample groups**
   48 backbones and 1,040 pendant groups



2. **Generate polymers**
   homopolymer repeat units



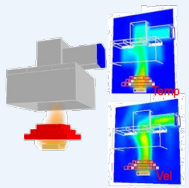
**Dataset v1**
1.7M homopolymer structures

3. **Correlate physical quantities**
   10 properties: thermophysical, degradation, combustion, etc.

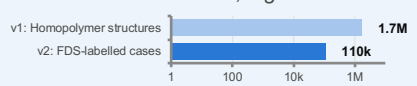
$$P = \frac{\sum_i N_i p_i}{\sum_i N_i M_i}$$

$P$: a polymer property
$p_i$: group-$i$ contribution to $P$
$M_i$: group-$i$ molar mass
$N_i$: group-$i$ occurance

4. **Fire Dynamics Simulator**
   Under different thicknesses and fluxes



**Dataset v2**
110k FDS-labelled data points over 55k homopolymers

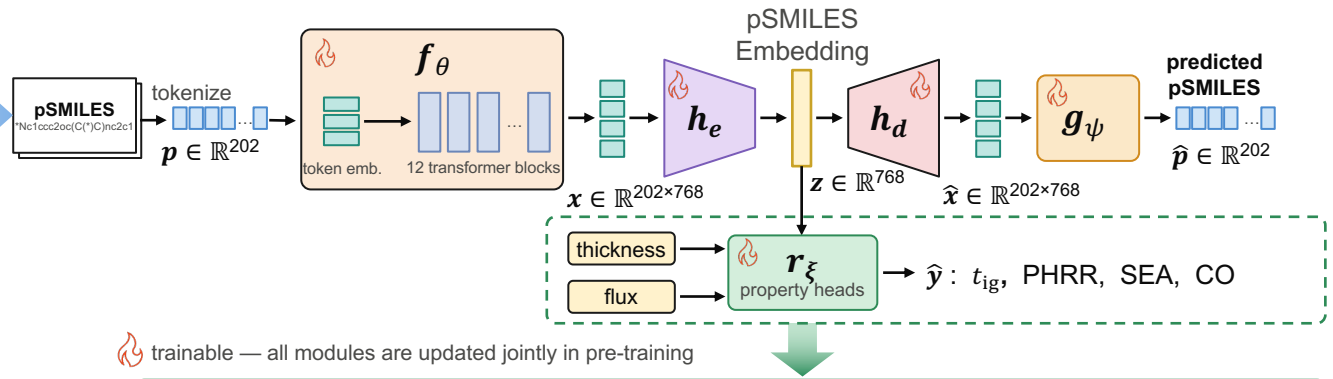



pSMILES strings

## b. Physics in the objective

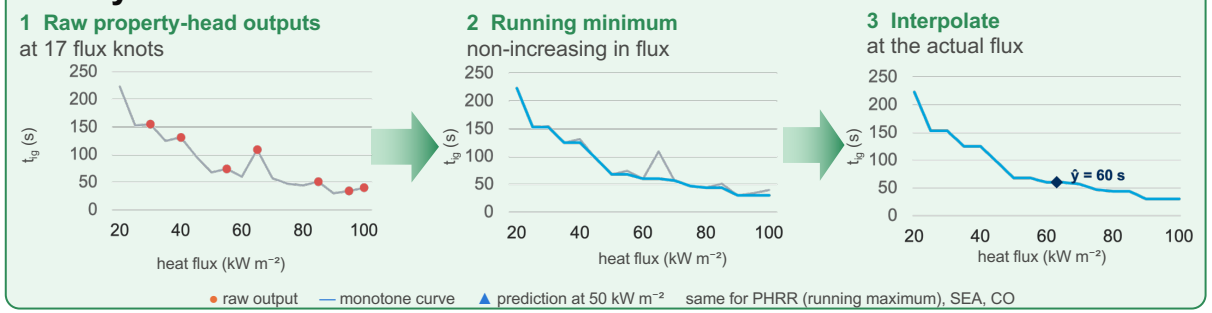
**Joint objective:** $\mathcal{L}_{\text{obj}} = \mathcal{L}_{\text{cls}}(\boldsymbol{p}, \hat{\boldsymbol{p}}) + \lambda_1 \mathcal{L}_{\text{rec}}(\boldsymbol{x}, \hat{\boldsymbol{x}}) + \lambda_2 \mathcal{L}_{\text{pred}}(\boldsymbol{y}, \hat{\boldsymbol{y}})$

**Property prediction regularizes the latent space: $z$ is organized by fire response as well as by structure.**

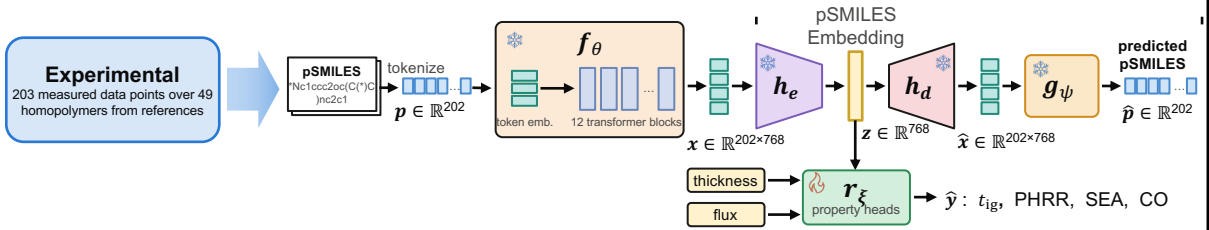



trainable — all modules are updated jointly in pre-training

## c. Physics in the architecture




# Fine-tuning for Property Prediction:




frozen — $f_\theta$, $h_e$, $h_d$ and $g_\psi$ keep their pre-trained weights

trainable — only the property heads $r_\xi$ are updated on the experimental data

# Latent Space Inverse Design:

